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Value from technology licensing – The role of monitoring and licensing experience

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ABSTRACT

Technology licensing is an important form of exchange in business-to-business markets. Licensees consider licensing technologies as a key mechanism to expand their technological asset base and to innovate. Even though exchange is the central focus of marketing academia, technology licensing contracts as a form of exchange have been largely overlooked. We take some steps to fill this void by studying the role of governance in licensing contracts. We rely on governance and specifically agency theory to study two-sided moral hazard problems in technology licensing. The two-sided nature of the problem is reflected in the presence of both upstream monitoring (i.e., the licensee monitoring the licensor) and downstream monitoring (i.e., the licensee being monitored by the licensor). We demonstrate that the effects of monitoring on licensee value creation are conditional on the licensee's experience in licensing. Specifically, we observe that, conditional on the licensee having accumulated a high level of licensing experience, upstream process monitoring and downstream output monitoring increase licensee market value while downstream process monitoring reduces licensee market value. Interestingly, we find no support for similar moderating effects of partner-specific experience. We rely on a unique database of original licensing contracts, historical data on past licensing contracts, and stock price data to analyze the value created by individual licensing contracts. The combined results may help licensees draft more valuable contracts.

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1. Introduction

For firms in technology intensive industries, expanding the technological asset base is of paramount importance to innovate and ensure future revenue generation. An important mechanism to access external technological inventions is technology licensing (Gupta, Smith, & Shalley, 2006). Technology licensing is a form of exchange where a licensee acquires technology from a licensor for the purpose of commercialization. For example, Pixium Technologies licensed technology from Stanford University to commercialize the replacement of light-sensitive cells to restore eyesight while Sanofi Pasteur signed a licensing agreement for flu vaccine technology with SK Chemicals to develop more effective vaccines. Technology licenses such as these give firms access to technological advances with commercial potential that cannot be developed (efficiently) in-house. As a consequence, licensing technologies accounts for a significant percentage of firms' R&D expenditures (e.g., 20% in the UK, Enterprise Research Center, 2016).

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Despite the benefits and prevalence of technology licensing, the ground realities show a chequered history of value creation. While [Anand and Khanna \(2000\)](#) demonstrate that, on average, licensing agreements generate larger excess returns than joint ventures, they also report that nearly half of technology licensing agreements fail to create positive value. Our main research objective is to explain this variation in value creation (i.e., abnormal stock returns). In view of the importance of licensing technologies to guard against obsolescence and to access new commercial opportunities, we take a licensee perspective and focus on value creation for licensees. Licensing is no guarantee for increased licensee firm value due to the inherent technological uncertainty and partner (i.e., licensor) uncertainty. A central premise of our research is that financial investors assess licensing agreements that govern the licensing exchange.¹ As US regulatory authorities require the disclosure of contracts that may potentially impact an investor's decisions ("material contracts"), investors can, upon obtaining this information via SEC filings, react by buying (selling) shares.

Within the realm of governance, we focus on monitoring. [Wathne and Heide \(2000\)](#) identified four core governance mechanisms: monitoring, incentives, selection, and socialization. They point out that ex-ante mechanisms such as contractually specified incentives and partner selection are ineffective in volatile environments (e.g. because of technological change) (p.45–46), which led other marketing scholars to call for more attention to ex post governance mechanisms ([Mooi & Ghosh, 2010](#)). While we control for royalties or upfront fees (cf. [Kotha, Crama, & Kim, 2018](#)) and the use of investments (cf. [Wathne, Heide, Mooi, & Kumar, 2018](#)), we assert that monitoring likely plays a significantly more important role in the valuation of technology licensing contracts. A report by the World Intellectual Property Organization ([WIPO 2016](#)) includes explicit advice that monitoring should be deployed to address uncertainty but fails to provide specific guidance. We address this gap and distinguish between different types and directions of monitoring. Further, the [WIPO \(2016\)](#) concludes that "each licensing negotiation is unique and it is difficult to apply the experiences of others or theoretical rules to the distinct situation at hand." We challenge this by demonstrating that accumulated licensing experience matters as experienced licensees reap significantly higher (and in one interesting instance, lower) financial benefits from monitoring. We also incorporate the role of socialization in our conceptual framework, as manifested in the licensee's prior licensing agreements with the same licensor.

Technology licensing requires significant investments but also creates market opportunities for licensees. These (potential) cash flows are of interest to investors who may evaluate licensing agreements depending on the mechanisms that are deployed to ensure control and facilitate coordination. Monitoring is a key governance mechanism in licensing exchange as it serves a dual governance role, as a control mechanism and as a coordination device. From a control point of view, monitoring enables a party to observe and address transgressions ([Kashyap, Antia, & Frazier, 2012](#)). Monitoring can thus protect the monitoring party from partner uncertainty via the exercise of authority ([Mellewigt, Madhok, & Weibel, 2007](#)). Monitoring also serves as a coordination mechanism ([Gulati, Lawrence, & Puranam, 2005](#); [Mellewigt et al., 2007](#)). Since monitoring requires information transfer, it can trigger further collaborative activities and reciprocal information exchange ([Mayer & Argyres, 2004](#); [Ryall & Sampson, 2006](#); [Wuyts & Geyskens, 2005](#)). Monitoring can thus alleviate technological uncertainty by bridging information asymmetries.

We contribute to the marketing literature in four ways. First, despite the significance of technology licensing, its ability to expand the licensee's technological asset base, its key role in innovation and the commercialization of inventions, and marketing's fundamental focus on exchange, it is surprising that technology licensing has been largely overlooked in marketing. Exceptions include [Wuyts and Dutta \(2008\)](#) who examined the impact of information disclosure and network position on dedicated technology firms' number of licensing agreements and [Jiang and Shi \(2018\)](#) who developed an analytical model to examine the consequences of inter-competitor licensing. In management, [Reddi et al. \(2018\)](#) examined partner uncertainty. We highlight a different but critical aspect of the multifaceted licensing phenomenon: the impact of monitoring in licensing contracts on licensee firm value.

Second, we contribute to the governance literature by acknowledging the double-sided moral hazard problem that is inherent to licensing agreements. Both the licensor and the licensee face significant partner uncertainty, as each partner can shirk on its obligations. To address this, licensing contracts include both upstream monitoring (licensee as the monitoring party) and downstream monitoring (licensee as the party being monitored). The licensee can monitor the licensor's progress in technology development as technologies are often licensed before they are fully developed while at the same time the licensee may be monitored by the licensor to ensure that it respects commercialization schedules. Academics rarely distinguish between upstream and downstream monitoring (see [Appendix A](#)), yet upstream and downstream monitoring are prevalent in managerial practice. Insight into their, potentially differential, effects may thus help advance the governance literature.

Third, we examine the role of experience as a boundary condition on the value creation implications of monitoring. We consider two types of experience: licensing experience and partner-specific experience. Licensing experience refers to experience which the licensee has accumulated via past licensing agreements with *any* licensor. Surprisingly, licensing experience, or more generally contracting experience, has received very little attention in the past literature. The studies by [Anand and Khanna \(2000\)](#) and [Reddi et al. \(2018\)](#) are notable exceptions, yet they were inconclusive regarding the performance consequences of contracting experience, possibly because they did not include interactions with governance. Partner-

¹ We verified that material licensing agreements are assessed carefully by investors and that specifically monitoring is essential to determine a licensing contract's value to the licensee, via interviews with industry experts, including an experienced IP lawyer and a Senior Technology Licensing Officer at a major northeastern university in the U.S.

specific experience (sometimes referred to as *repeated partnering*) refers to the licensee's previous experience with the same licensor. To the best of our knowledge, no prior work in marketing has made this explicit distinction between contracting experience and partner-specific experience, which constitutes another contribution to the field.

Fourth, we draw on secondary data sources to examine the impact of monitoring on licensee market value. From the Contracting & Organizations Research Institute (CORI) database we extract the actual (written) content of technology licensing contracts and from the DataStream database, we extract data on firm financial value. Only a few studies have examined the linkage between actual licensing contracts and firm performance (Jayachandran, Kaufman, Kumar, & Hewett, 2013). We are among the first to demonstrate that governance efforts at the transactional level explain firm value (cf. Weber, Mayer, & Rui, 2009), thus elevating the importance of transactional governance. Our study of actual contracts, rather than self-reported measures, and their impact on licensee market value (as opposed to intermediate constructs such as opportunism) are unique features of this research project. Finally, we use secondary data from SDC Platinum to extract all prior licensing agreements and thus have unique access to prior interactions (instead of self-reported measures which may be subject to reporting biases) to study the moderating effects of licensing and partner-specific experience.

We find that, conditional on the licensee having accumulated a high level of licensing experience, upstream process monitoring and downstream output monitoring increase licensee market value while downstream process monitoring reduces licensee market value. These findings have several important implications. First, more monitoring is not necessarily better as downstream process monitoring harms the financial outcomes for experienced licensees. Second, these findings underscore the academic and managerial importance of distinguishing between different forms of monitoring (process versus output; upstream versus downstream). Third, the findings also show a clear pattern for experience as a boundary condition: the effects of different types of monitoring on licensee firm value manifest themselves when licensees are experienced in terms of licensing, but largely disappear for inexperienced licensees. Interestingly, we do not find similar effects for partner-specific experience, pointing to an important difference between both forms of experience. This finding is important as prior research has predominantly studied partner-specific experience (e.g. Argyres & Mayer 2007; Vanneste & Puranam, 2010), leaving general experience with any partner largely unexplored. Yet, in the context of technology licensing, licensing experience appears to dominate partner-specific experience in determining the financial outcomes of monitoring in licensing contracts. Our actionable recommendations may help licensees make appropriate governance decisions.

2. Theory

While various forms of licensing exist, (e.g., brand licensing, see Jayachandran et al., 2013), the focus of this paper is on technology licensing (e.g., Arora, Fosfuri, & Rønde, 2013; Somaya et al., 2010; Wuyts & Dutta, 2008). Technology licensing involves a licensor and a licensee where the licensor provides the licensee with access to technology, often in the form of blueprints, patents, and know-how in exchange for either royalty or fixed-fee payments. The licensee interacts with the licensor to cope with the complexity of technology exchange and facilitate knowledge absorption, implementation, and commercialization. Subsequent sales of innovations based on the licensed technology generate cash flows for the licensee.² In sum, technology licensing is a prototypical example of the classic double-sided moral hazard problem, alluded to by Agrawal and Lal (1995), Gallini and Lutz (1992), and Lafontaine (1992). The uncertainties faced by the licensees include (1) unpredictability inherent to the technology that is subject to the license (*technological uncertainty*) and (2) unpredictability regarding the licensor's behaviors and intentions (*partner uncertainty*). We discuss each in turn.

Technological uncertainty is the inherent uncertainty regarding the technology's value and applicability (Wuyts & Dutta, 2008) and stems from the complex linkages between technological components and the unpredictability of technological progress. For example, Roche Molecular Technologies licenses out a polymerase chain reaction (PCR) technology for a wide range of applications, including DNA matching for crime suspects, authentication of types of food (e.g., to determine real caviar), and determination of patients' responses to medicine. Although Roche developed and licenses out the basic technology, each subsequent application requires substantial additional development and is highly complex, requiring up- and downstream communication and coordination between Roche and its licensees.

Partner uncertainty arises due to information asymmetry, as the licensor often has an information advantage over the licensee regarding the technology's value. Such information asymmetry is inherent to technology licensing as the goal of the exchange is to acquire intellectual property the licensee does not yet possess. Hence, licensing exchange partners may face moral hazard or hidden action problems. For example, significant *ex post* transfer problems may arise when the licensee "sits" on the license to protect its own technology (John, Weiss, & Dutta, 1999). This practice, termed "shelving", where the licensee acquires the rights to a technology with the explicit intention to stop it from entering the market or as a delaying tactic (Dechenaux, Thursby, & Thursby, 2009; Palomeras, 2007), is a source of partner uncertainty for the licensor. Both parties may face partner uncertainty, caused by moral hazard or hidden action problems, necessitating two-way communication and control *ex post* to address these concerns.

² Compensation via other technology is called cross-licensing and falls outside of the scope of this study.

2.1. Bright and dark sides of monitoring

To deal with these uncertainties, licensees and licensors purposely incorporate monitoring in licensing contracts. Yet, prudence in the specification of monitoring is required, as past work has demonstrated that more monitoring is not necessarily better (Heide, Wathne, & Rokkan, 2007; Kashyap et al., 2012). Clearly, monitoring possesses “bright” and “dark” sides that differentially impact outcomes.

On the “bright side”, the economics literature considers contracts as tools that mitigate inefficiencies created by conflicting incentives (Eisenhardt, 1989; Williamson, 1985). Accordingly, monitoring serves a control function and enables the monitoring party to ascertain that desired actions are undertaken, and performance levels are reached, while minimizing defiant behavior or deficient output (Heide, 1994). This may involve discussing negative behaviors. Strategy scholars also discussed contracts as coordination devices that blueprint roles and responsibilities of both parties to the exchange (Macaulay, 1963; Mayer & Argyres, 2004). Mellewigt et al. (2007) note specifically that monitoring fosters coordination as it involves regular and ongoing communication between the licensor and licensee. According to legal scholars (Gilson, Sabel, & Scott, 2010), monitoring is of particular relevance to high-tech industries where partner and technological uncertainty are both high and persistent during the exchange. Communication flowing from monitoring enhances predictability and thereby, to some degree, reduces partner uncertainty. Communication also facilitates the transfer of implicit knowledge, which may help reduce technological uncertainty.

On the “dark side”, monitoring may be perceived as an obtrusive form of control by the receiving party. Monitoring may offend the monitored party's sense of autonomy, which can cause “reactance” (Brehm, 1966; Heide et al., 2007). Reactance refers to the monitored party's attempts to reassert its autonomy, undermining the social component of the exchange and reducing both the effectiveness of enforcement and the willingness of the involved parties to communicate and exchange knowledge. Reactance generates self-serving behaviors that work counterproductively in terms of performance (Kashyap et al., 2012).

Concluding, we assert that prespecified monitoring *signals* to financial investors that the licensee may gain control and coordination benefits or that value creation may be constrained by reactance and self-serving behaviors. Investors' reading of these signals, in terms of “bright” or “dark” side effects, drives their expectations of future returns and, thus, licensee market value.

2.2. Types of monitoring

Following Heide et al. (2007) and Crosno and Brown (2015), we distinguish between process and output monitoring. *Process monitoring* refers to the measurement or “metering” of actions undertaken by an exchange partner to create outputs. *Output monitoring* refers to the situation where the results of the partner's actions are metered. Several studies (see Appendix A) have distinguished between output and process monitoring and pointed to their theoretical differences (Grewal, Kumar, Mallapragada, & Saini, 2013; Heide et al., 2007; Kashyap et al., 2012). We contribute to this literature by considering that monitoring may be upstream and downstream. Downstream monitoring targets the licensee while upstream monitoring targets the licensor. Appendix A shows that previous studies typically examine downstream or (to a much smaller extent) upstream monitoring—only Dahlquist and Griffith (2014) consider both. We argue that it is important to incorporate both, as both may contribute to value creation, yet their effects likely differ (as we explain subsequently). Distinguishing between output and process monitoring and between upstream and downstream monitoring is unique to the literature. To illustrate how each form of monitoring can manifest itself, we provide an example of each in Appendix B.

Taking the perspective of the licensee, we develop arguments on the impact of different forms of monitoring on licensee firm value. It is important to note that while process monitoring has theoretical relevance (and is observed empirically) both downstream and upstream, output monitoring only has empirical relevance downstream. The key “output” expected from the licensor consists of the authorization for the licensee to utilize the licensed property, which is at the core of any licensing agreement. Financial outputs are not expected from a licensor. While the licensor may be expected to complete technology development, upstream monitoring of technology development outputs is unlikely for two key reasons. First, the technological uncertainty that is inherent to technology development makes it difficult (if not impossible) to contractually specify all possible contingencies (Crocker & Reynolds, 1993). Second, the information asymmetry that is at the source of licensing contracts amplifies technological uncertainty and introduces partner uncertainty, further limiting the licensee's ability to contractually specify (technology) development outcomes. In sum, upstream output monitoring is unlikely to be observed in licensing contracts³.

Thus, we focus on three forms of monitoring: upstream process monitoring, downstream process monitoring, and downstream output monitoring. We refrain from deriving formal main effect hypotheses of monitoring on licensee firm value as past work (e.g. Heide et al., 2007) has documented both positive and negative effects on exchange performance. Instead, in this section we briefly review arguments regarding these main effects and in the next section we derive a set of novel hypotheses on the moderating effects of two sources of experience.

³ This is also borne out by our empirical findings. The licensing contracts in our sample do not include upstream output monitoring.

First, upstream process monitoring entails the licensee monitoring the licensor's remaining development and production processes. We expect that upstream process monitoring serves an important coordination role, facilitating communication between licensee and licensor. As the licensee needs information from the licensor to convert the licensed technology to a product (Somaya, Kim, & Vonortas, 2010), it benefits from the institutionalization of communication on technology development. Even when the licensed technology is clearly described in patents or blueprints, additional knowledge transfer is often required to set up production for the final product. Upstream process monitoring provides an opportunity for licensees to coordinate information exchange, increasing the probability of reaching a desired outcome (and avoiding undesired outcomes) when compared to a licensing agreement without such coordination.

Second, downstream process monitoring occurs when the licensee's commercialization processes are monitored by the licensor. Past work by Arora and Gambardella (2010) shows how licensors are motivated by their desire to quickly recoup investment costs, as opposed to maximizing returns for the licensee. As a result, licensor and licensee interests are often misaligned and monitoring by the licensor may be poorly directed. The licensor is also further removed from the end market, increasing the odds of misalignment with market requirements, complicating and delaying decision-making (Lee, Padmanabhan, & Whang, 1997). In sum—and opposed to upstream process monitoring which provides the licensee with (potential) access to additional knowledge on the licensed technology—downstream process monitoring may work counterproductively and the licensor's inputs on commercialization processes may be misguided and constraining. The licensee may perceive such monitoring as autonomy-depriving and obtrusive, triggering reactance (e.g., Brehm, 1966). Downstream process monitoring also does not provide particular information benefits to the licensor beyond the assurance of the licensee investing the necessary time and resources to commercialize the technology (as opposed to shelving it). In sum, downstream process monitoring has limited informational or coordination benefits, but rather appears intended to keep licensees on their toes, which may trigger negative licensee reactions.

Third, downstream output monitoring refers to the licensor's right to monitor the licensee's financial outcomes. While process monitoring focuses on observing behaviors and intermediate steps, output monitoring entails the specification of end goals and gives the monitored party full discretion as to how to reach these goals. As such, we expect that the “dark side” of monitoring does not apply as the specification of targets works as a motivator for the monitored party, and unlikely triggers negative feelings and reactance (Heide et al., 2007). Prior research has suggested that output monitoring is not obtrusive since it does not constrain the monitored party in terms of processes and activities to reach set targets (Anderson & Oliver, 1987). The specification of targets motivates the monitored party to exert extra efforts to reach these targets, stimulates compliance, and reduces opportunistic or other wasteful behaviors (Kashyap et al., 2012). The effectiveness of output monitoring is in part due to the verifiability of “hard” performance metrics, which are useful when licensors face an adverse selection problem *ex-ante*. Some licensees may plan to shelve acquired technologies to promote their own technologies, keep competitors from entering the market, or avoid cannibalization (Dechenaux et al., 2009). Downstream outcome monitoring can prevent this as ill-motivated licensees would be reluctant to deal with licensors that seek to include financial monitoring. Since financial outputs cannot be realized when the licensed technology is shelved, the inclusion of downstream output monitoring serves as a credible signal of good intent to investors (Kirmani & Rao, 2000).

2.3. The moderating role of experience

As noted in the introduction, the WIPO (2016) questions if learning can occur in negotiating licensing contracts and if generalizable insights can be derived that apply across distinct contracts. This raises an interesting question, whether licensees can rely on their past experiences to better reap the financial benefits of licensing exchange. Casual observation shows that firms, such as Roche Pharmaceuticals repeatedly strike technology licensing agreements, both with the same and different partners—two potentially relevant sources of experience.

In our theorizing, we attribute a key role to the licensee's experience with past licensing agreements. The prior literature has overwhelmingly focused on partner-specific experience, i.e. experience accumulated via prior licensing agreements with the *same* partner (e.g., Holloway & Parmigiani, 2016; Wuyts & Geyskens, 2005). We identify an additional and more general source of experience, licensing experience, i.e. experience accumulated via prior licensing agreements with *any* partner. We expect that in technology licensing, where subsequent transactions cover different technologies and technological evolution renders prior technologies obsolete, partner-specific experience may be less impactful than general licensing experience. The moderating effects of partner-specific experience and licensing experience rely on different theoretical mechanisms. The “received wisdom” in marketing and strategy literatures, is that partner-specific experience helps exchange partners develop relationship-specific heuristics and norms that facilitate coordination and foster collaboration (Uzzi, 1997; Heide, 1994). While these routines and relational norms are not portable to other relationships, they may improve joint problem-solving in new transactions between the same two parties. An important difference between relational norms and monitoring, relevant for the development of hypotheses concerning the moderating role of partner-specific experience, is that monitoring relies on a logic of consequences (i.e., actions are driven by subjective assessments of outcomes of alternative courses of action) whereas relational norms rely on a logic of appropriateness (i.e., actions are shaped by rules relevant to the current situation) (Schulz, 2014; Heide & Wathne 2006).

Licensing experience, in contrast, enables the licensee to compare between different technologies *and* partners. Successful as well as unsuccessful prior experiences with different licensees help build higher-order licensing routines and capabilities to effectively design and deploy governance structures (Vanneste & Puranam, 2010). Lewin, Massini, and Peeters (2011)

explain that cumulative experiences lead to the development of routines or behavioral regularities, which are the building blocks of capabilities. Of interest are so-called higher-order routines, which transcend the individual transaction and are portable to new situations (e.g., routines for specifying and aligning objectives). It is precisely the portability of experiences that leads us to expect that licensing experience dominates partner-specific experience in licensing exchange. In the method section, we will examine the explanatory power of both sets of interactions (with partner-specific vs. licensing experience, respectively).

3. Hypotheses

Having discussed types of monitoring and experience, we now proceed to hypothesis development. We organize the argumentation for the hypotheses around (a) control effects, (b) coordination effects, and (c) reactance effects associated with monitoring and how these interact with licensing experience and partner-specific experience. Table 1 summarizes the underlying logic for the hypotheses, while Fig. 1 presents the conceptual framework.

3.1. Upstream process monitoring

We start with partner-specific experience. The existence of prior licensing agreements between the licensee and the licensor suggests the creation of relationship-specific heuristics and norms (Uzzi, 1997), which obviates the need for deploying additional control mechanisms as well as for institutionalizing communication flows within the new licensing agreement. When norms of information sharing arise, exchange partners engage in more frequent and bidirectional communication (Mohr & Sohi, 1995). Upstream process monitoring may then be less valuable as a vehicle for information exchange. More problematically, once information sharing norms have been created the licensor may perceive the inclusion of upstream monitoring as a “defection” from past norms in that, *ceteris paribus*, greater control is applied. Such “defections” may signal utility-maximizing behavior (Heide & Wathne, 2006). As a result, the licensor may start questioning the licensee's motives and perceive such monitoring as unwarranted and excessive. The logic of appropriateness on which relational norms rely conflicts with the logic of consequences of the monitoring mechanism (Heide & Wathne, 2006). In sum, the dark side of upstream process monitoring may chip away from the bright side in the presence of more partner-specific experience. Therefore, we expect that:

H1a: Upstream process monitoring *increases* licensee market value *less* when the licensee has accumulated *more* partner-specific experience.

We now turn to the effect of licensing experience. We expect that licensees that have accumulated greater prior licensing experience are better able to enjoy the salutary effects of upstream process monitoring. Information asymmetry, intrinsic to a licensing agreement, makes it difficult for the licensee to monitor the licensor's development processes and upstream process monitoring can help stimulate information exchange. More experienced licensees are familiar with the ground realities of monitoring, including the articulation of what is expected from the licensor and effective approaches to measuring and observing. Inexperienced licensees, in contrast, may have great difficulty to effectively deploy the communication channels that are formalized by monitoring. In sum, while upstream process monitoring provides licensees with an *opportunity* to coordinate with the licensor and institutionalize technology-related communication flows, experience from past licensing likely facilitates taking advantage of the opportunity for value creation.

Even though process controls are sometimes perceived as “autonomy-depriving, unwarranted interventions” (Grewal et al., 2013, p.381), the licensor benefits directly from upstream process monitoring. It is in the licensor's best interest that the licensee is fully informed on the technological developments to commercialize the invention successfully as successful commercialization is likely to increase royalty payments. Because of this direct economic benefit to the licensor, we do not expect that upstream process monitoring triggers reactance from the licensor or negative feelings of being subject to obtrusive control. In sum, we expect that:

H1b: Upstream process monitoring *increases* licensee market value *more* when the licensee has accumulated *more* licensing experience.

3.2. Downstream process monitoring

We expect that downstream process monitoring reduces licensee market value when the licensee has accumulated more prior experience with the same licensor. On the one hand, licensees do not gain control benefits from downstream monitoring as they are the monitored party, and downstream process monitoring also does not offer the licensee significant coordination benefits as the licensee has little to gain from sharing information on market commercialization processes. On the other hand, the licensor's use of downstream process monitoring as a control mechanism signals a logic of consequences, which contrasts with the logic of appropriateness associated with socialization resulting from preexisting ties between the licensee and licensor (Heide & Wathne, 2006). We expect that this mismatch of utilizing downstream process monitoring in

Table 1
Moderation Effects of Partner-specific vs. Licensing Experience and Logic of Hypotheses.

Hypothesized Moderating effect	Hyp	Control effect of monitoring	Coordination effect of monitoring	Reactance effect of monitoring	Total effect
Upstream process mon. × Partner-specific exp.	H1a	Weak: Information asymmetry hinders upstream process control; Social norms also serve control role	Weak: Socialization stimulates sharing information	Strong: Conflict between logic of consequences vs logic of appropriateness	–
Upstream process mon. × Licensing exp.	H1b	Weak: Licensing experience does not bridge the information asymmetry that hinders upstream control	Strong: Higher ability to articulate expectations and take advantage of sharing information on technology development	Weak: Licensor benefits directly from improved information sharing	+
Downstream process mon. × Partner-specific exp.	H2a	Weak: The licensee is the controlled party, no benefits from having processes controlled	Weak: No immediate benefits for licensee from sharing information on commercialization	Strong: Conflict between logic of consequences vs logic of appropriateness	–
Downstream process mon. × Licensing exp.	H2b	Weak: The licensee is the controlled party, no benefits from having processes controlled	Weak: No immediate benefits for licensee from sharing information on commercialization	Strong: An experienced licensee may perceive downstream process controls as misdirected, autonomy-depriving.	–
Downstream output mon. × Partner-specific exp.	H3a	Weak: The licensee is the controlled party, no benefits from having processes controlled; social norms also serve control role	Strong: partner-specific experience facilitates determination and internalization of outputs	Weak: output monitoring leaves the decision of how to reach the outputs to the licensee	+
Downstream output mon. × Licensing exp.	H3b	Weak: The licensee is the controlled party, no benefits from having processes controlled	Strong: experienced licensee has higher-order capability to decide how to reach set outputs	Weak: output monitoring leaves decision how to reach outputs to licensee	+

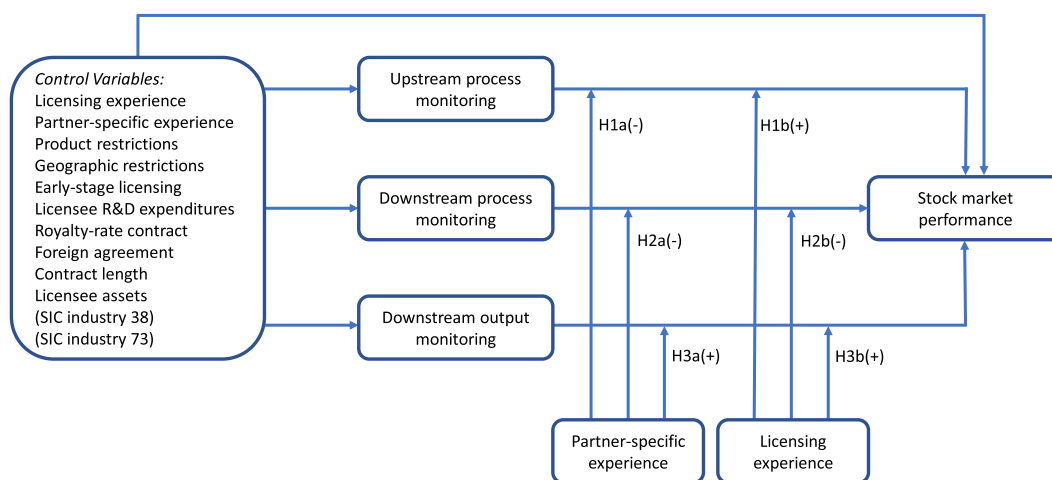


Fig. 1. Conceptual model.

case of partner-specific experience increases the risk of conflict and thereby negatively affects the licensee's market value. While the dark side of downstream process monitoring is likely to be more salient when the licensee and licensor have been involved in licensing agreements in the past, there is no bright side to compensate for this negative performance effect. The licensee is unlikely to learn from the licensor controlling its commercialization processes, for reasons outlined in the above theory section. We therefore expect:

H2a: Downstream process monitoring *decreases* licensee market value *more* when the licensee has accumulated *more* partner-specific experience.

A licensee's prior licensing experience may also cause monitoring to be perceived as unwarranted, another form of a dark side to downstream process monitoring. More experienced licensees, with greater accumulated knowledge on commercializing licensed technologies, are more likely hindered by external process control, consistent with theories of process control and motivation (Brehm & Brehm, 2013). They, thus, likely perceive downstream process monitoring as misdirected, autonomy-depriving, and unwarranted (Grewal et al., 2013). While downstream process monitoring inherently carries some risk of conflict, such conflict more likely materializes among licensees with more licensing experience. At the same time—and as argued earlier—downstream process monitoring unlikely offers the licensee significant control or coordination benefits. As a consequence, investors likely react negatively to downstream process controls when the licensee has accumulated more experience via past licensing agreements. We expect that this contingency argument will be manifested through a negative interaction effect between process monitoring and licensee experience on licensee market value:

H2b: Downstream process monitoring *decreases* licensee market value *more* when the licensee has accumulated *more* licensing experience.

3.3. Downstream output monitoring

When the licensee has partner-specific experience, via prior licensing contracts with the same licensor, the effect of downstream output monitoring on licensee market value may be strengthened. Partner-specific experience stimulates information exchange and the internalization of the other party's goals (Heide & Wathne, 2006). Hence, partner-specific experience facilitates the determination of commonly agreed output targets and increases the licensee's motivation to internalize and pursue those targets. In addition to these coordination benefits, and unlike in hypotheses H1a and H2a where we expected a mismatch between the control afforded by monitoring and prior exchange norms, we do not expect a mismatch here as output monitoring leaves the decision of how to reach the outputs to the licensee. The specification of agreed-upon outputs aligns objectives and works performance-enhancing (Bello & Gilliland, 1997). Hence:

H3a: Downstream output monitoring *increases* licensee market value *more* when the licensee has *more* partner-specific experience.

We expect that the positive effect of downstream outcome monitoring is also stronger when the licensee has accumulated prior licensing experience. The increased efforts to obtain certain commercial goals more likely result in increased market value for those licensees that have more experience with commercializing new technologies. Output monitoring gives

licensees the managerial discretion to select the best approach to reach desired outcomes (Finkelstein & Hambrick, 1990). An implicit assumption to using such discretion productively is that the licensee has the necessary experience that enables it to identify strategic alternatives and select the best approach. We expect more experienced licensees to be better equipped to identify and select alternatives and thereby to achieve desirable outcomes. Previous studies have shown that contracting over time fosters learning, which improves coordination within new contracts (Vanneste & Puranam, 2010). Thus, when the licensee is more experienced in licensing, incorporating outcome monitoring in the licensing contract may be perceived as a stronger signal of intent toward investors. In summary, we expect that the positive effect of downstream outcome monitoring on licensee market value is stronger when the licensee has accumulated more licensing experience via past licensing agreements:

H3b: Downstream output monitoring *increases* licensee market value *more* when the licensee has *more* licensing experience.

4. Method

4.1. Context

We study the effects of technology licenses on licensee market value in high-technology markets. We focus on high-technology industries in SIC codes 28 (chemicals and allied products), 38 (instruments and devices), and 73 (business services, dominated by software) as technology licensing is almost exclusively observed in these industries.

We draw our data from various sources. Using the DataStream database, we collect stock market data. The US Securities and Exchange Commission (SEC) requires listed firms, to which our sample is constrained, to disclose material contracts as part of their 8-K, 10-K, 10-Q, and S-1 filings. The University of Missouri's Contracting and Organizations Research Institute (CORI) has kept track of SEC filings (retrieved from EDGAR). The benefit of the unique CORI database is that contracts are categorized along several dimensions such as the type of agreement and industry. In addition, Anand and Khanna (2000) state that 'SDC data on the date of the event are misstated in many instances', creating a major obstacle for event studies. CORI staff verifies the announcement date of each licensing agreement via a careful search and, where required, direct contact with the firms involved. An accurate announcement date is essential to our market valuation model which relies on identifying a narrow window around the event. Finally, while many agreements listed in SEC's EDGAR database are qualitatively different from technology licensing agreements (e.g., debt agreements), CORI carefully verifies and reports whether an agreement is an *actual* licensing agreement.

Since CORI collected licensing contracts only up to the end of 2006, we selected all technology licenses from the aforementioned SIC industry codes from 1/1/2003 to 12/31/2006.⁴ This specific time period allows for a good test of the developed theory. The sampling period was relatively stable and predates the global financial crisis, the aftermath of which reverberated for many years. Importantly, the phenomenon of technology licensing and the theoretical arguments related to monitoring and experience are not likely time-specific.⁵

We carefully went through the description and contents of all technology licenses and excluded cross-licenses, agreements with the sole purpose to obtain licenses as collateral for loans, re-negotiations, and licenses with more than two parties to the exchange. This resulted in 204 licensing agreements. Because our dependent variable requires us to observe variation in stock prices, we only included licensing firms for which these could be retrieved from DataStream, reducing the sample size to 136.⁶ From the Thomson SDC database, we then extracted the number of prior licenses engaged in by the licensees included in our sample (no observations were lost in this process). Our total data collection efforts thus resulted in 136 licensing agreements for which we had access to the original contracts (required to identify and measure monitoring), announcement dates, licensee stock market data, licensing experience, and a series of control variables. In what follows, we describe the measures of our dependent and independent variables.

4.2. Dependent variables

Our dependent variable is the change in the licensees' stock market value in response to the licensing announcement. Since technology licensing agreements are discrete stand-alone events, changes to stock returns provide a good metric for assessing their financial value (Warren & Sorescu, 2017). The change in stock market value is a forward-looking measure that

⁴ We note that due to funding shortfalls, CORI was unable to collect as many agreements in 2004 as for the other years.

⁵ Upon close examination of industry reports, we did not come across any indications that anything substantial has changed over the past fifteen years in terms of how contracts are negotiated or shaped. On the contrary, as a 2016 report on "Successful Technology Licensing" from the World Intellectual Property Organization clearly illustrates, issues related to specifications, deliverables, control, and leverage are still primary concerns in technology licensing. We also reached out to seasoned industry experts who unanimously confirmed that these are "evergreen" issues and that the fundamentals of technology licensing contracts and the role of monitoring therein remained largely the same over time. In sum, the precise time period of data collection does not appear to be an issue.

⁶ We tested for mean differences between the sample of 204 and 136 responses on variables for which we have full information. Specifically, we tested for mean differences in investments ($t = 0.6472, p > .05$), geographical restrictions ($t = 0.3517, p > .05$), early-stage licensing ($t = 0.7054, p > .05$) and whether the contract was a royalty contract ($t = 0.9536, p > .05$). Taken together, these tests do not indicate any selection issues *prima facie*.

reflects investors' consensus forecast of the change in the long-term value of the firm attributable to the event (Gielens, Van de Gucht, Steenkamp, & Dekimpe, 2008) and is particularly relevant given agency theory's focus on maximizing shareholder payoffs (cf. Kim & Mahoney, 2005). Investors react to the announcement by buying (selling) stocks, which positively (negatively) affects the firm's stock price. Investors can obtain precise details on licensing agreements through the Security and Exchange Commission's (SEC) EDGAR databases (which is also the source of CORI). The SEC requires firms to disclose licensing contracts if, and when, there is an expectation that such agreements potentially influence the market value of the firms involved. Our forward-looking measure is useful because it may take one or more years for licensing to generate sales revenues and because (historical) or broad-based performance measures are a composite of many actions of a particular firm whereas stock price reactions can be attributed to the isolated event under study, even if such an event is at the "transactional" level.

We use the DataStream database to collect stock market information. To determine the event window, we acknowledge prior literature on market reactions to the announcement of licensing agreements. Anand and Khanna (2000) provided evidence of knowledge leakage on the preannouncement day (−1) and excess returns in the days following the announcement of a licensing agreement. We, therefore, considered a series of event windows that account for both knowledge leakage and knowledge dissemination. Following the approach proposed by Brown and Warner (1985) and applied in marketing (e.g. Gielens et al., 2008; Raassens, Wuyts, & Geyskens, 2012), we then determined the optimal event window empirically by calculating the Patell (1976) Z-statistic for the average cumulative standardized abnormal returns (CSAR) across alternative event windows. Table 2 reports the average CSAR and significance levels. We rely on the event window with the largest Patell Z-statistic, [−1, +3]. In this window, 50.40% of licensees experience negative abnormal returns while 49.60% of licensees experience positive returns.

4.3. Independent variables

Below, we present the measures for monitoring, licensing experience, along with measures for various control variables which we include in the specifications of the licensee market value and monitoring equations. Further details (measurement items, theoretical and data sources) are provided in Appendix B.

Upstream/downstream process monitoring is coded, using the original, written licensing contract. We used the following procedure. As a first step, based on a detailed reading of a subset of licensing contracts, we generated a comprehensive list of processes that are monitored in our sample of licensing agreements. Next, we trained 4 RAs in coding 20 licensing contracts based on the presence or absence of these processes, and we resolved any disagreements via discussion. Once convergence was reached among coders, the remaining contracts (116) were distributed among the team. Process monitoring was calculated by counting the number of processes subject to monitoring. While the number of monitoring clauses may serve as a proxy for the number of processes being monitored, the overlap between clauses and actual processes in our sample is imperfect and we therefore opted for the more direct measure, i.e. counting processes. Our coding approach largely mirrors Kashyap et al. (2012) coding approach. One exception is that due to the larger number of contracts in our sample, we jointly coded a subset of contracts and then distributed the remaining contracts among the coders once convergence was reached. As a result, we cannot report indices of agreement among coders. We code three process monitoring items (see Appendix B for an example of each): monitoring the development process, monitoring the production process, and monitoring marketing, sales, advertising, and communication processes. As the exchange of the technology may require further development, inputs in the setup of production processes, and inputs into marketing, sales, advertising, or communication processes by either party, we separately coded if such monitoring was undertaken by the licensor or the licensee.

Downstream output monitoring is also coded using the aforementioned process and measures if the licensing contract explicitly includes a paragraph on how the financial outputs of the technology license are monitored.

Licensing experience is measured by the number of prior licensing agreements obtained by the licensee and as reported in the SDC Platinum database between the first of January 1990 and the date of announcement of the focal licensing agreement.

Partner-specific experience is measured by the number of prior licensing agreements signed by the licensee with the same licensor partner as the focal agreement, and as reported in the SDC Platinum database between the first of January 1990 and the date of announcement of the focal licensing agreement.

Investments are coded by counting the number of categories specified in the contract, in which the licensee agrees to commit resources. The investment categories include R&D investments, manufacturing investments, and marketing investments. Higher values indicate a greater scope of investments agreed on, which may increase downstream process monitoring.

Product and geographic restrictions are commonly included in licensing contracts, reflecting the licensor's desire to limit the usage domain and/or geographic area in which the licensee is allowed to use the licensed technology. As technology may have different productive uses, product restrictions could allow the licensor to sell the technology for a different purpose to another licensee. Similarly, geographic restrictions enable the licensor to license out its technology to various licensees in different geographic regions. This may have consequences for the impact of the licensing contract on the licensee's market value. Based on the contract, we created two dummies to capture if the license includes product and geographical restrictions, respectively.

Early-stage licensing measures if the licensor is responsible for development or other aspects after the closing of the licensing agreement. This is an important variable as the resulting dependence of the licensee on the licensor may give rise to hold-

Table 2
Average Cumulative Standardized Abnormal Returns for Alternative Event Windows.

Window	Average Cumulative Standardized Abnormal Returns	Patell z-statistic
[−3, 1]	0.37	1.60
[−3,2]	0.45	1.78
[−3,3]	0.57	2.08
[−2,1]	0.41	1.97
[−2,2]	0.49	2.11
[−2,3]	0.61	2.40
[−1,1]	0.41	2.29
[−1,2]	0.49	2.36
[−1,3]	0.61	2.64

up problems, encouraging the licensee to demand upstream process monitoring and decreasing licensee market value. We coded this as a dummy variable, based on the licensing contract.

Licensee R&D expenditures are measured as R&D spending (in \$1000) by the licensee. This measure is taken from DataStream.

Royalty-rate contract measures if the licensee compensates the licensor via a royalty-rate (as opposed to fixed-fee), which likely decreases downstream process monitoring. Note that per SEC rules, parties to the licensing agreement may choose to redact out specific numbers such as the actual royalty percentage. As this practice is pursued overwhelmingly in the contracts deposited in the EDGAR database, the percentage itself could not be coded. Instead, this variable captures the starkly different roles of variable vs. fixed-fee incentives (Bajari & Tadelis, 2001).

Contract length is the (log-transformed) word count of the contract and serves as a proxy for the extensiveness or scope of the contract. Longer contracts feature more detail and could inhibit flexibility or curb potential opportunism or other undesirable behaviors, both of which could impact how analysts assess contracts.

Licensee assets are measured as the total (log-transformed) \$ financial assets of the licensee as reported by DataStream.

Finally, we use two industry dummies (SIC code 38—instruments and devices, and SIC code 73—business services, dominated by software) to absorb inter-industry differences. SIC code 28, chemicals and allied products, serves as the baseline industry.

Table 3 provides descriptive statistics for all variables, as well as the correlation matrix.

4.4. Model specification

Our model specification approach is informed by the need to account for four constraints imposed by our data structure, measures, and multiple equations. First, our sample size of 136 licenses is composed of 104 unique licensees. As a number of licensee firms are observed more than once in our sample, the assumption of independent observations may no longer be valid. Second, the three monitoring variables could potentially be endogenous with regard to licensee market value. Licensees may strategically strive to include monitoring when they expect monitoring to be beneficial for them in terms of value. Third, the four equations may be more efficiently estimated by taking into account correlated residuals. Fourth, our monitoring variables are respectively ordinally scaled (process) and binary scaled (output) scaled while licensee market value is scaled continuously, requiring us to “mix” differently scaled dependent variables.

To address these challenges, we specify a recursive set of equations and estimate these jointly, using Roodman (2008) Conditional Mixed Process (CMP) regression. CMP makes use of a simulated maximum likelihood-based seemingly unrelated regression (SUR) estimator to jointly estimate a recursive set of equations (See Kashyap et al., 2012; Antia, Zheng, & Frazier, 2013 for additional details on the method, and its application to marketing).

To address the first constraint, we specify a clustered robust estimator, clustered on the licensee, to account for potential non-independence.

To address the second concern of endogeneity, we leverage the option in CMP to include the residuals of the monitoring equations in the outcome equation (Cf. Kashyap et al., 2012). As such, our specification follows that of a control function approach as deployed commonly in marketing (e.g. Mooi & Ghosh, 2010). We used two industry dummies (to indicate SIC codes 38 and 73) as exclusion restrictions. Differences in industry practices can give rise to different contracting practices in terms of monitoring. We verified that such was the case empirically and note that in our final model each instrument correlates significantly with the monitoring variables ($p < .05$) suggesting relevance of the instrument. Traditionally, for continuous dependent variables, instrument strength can be assessed using the approach advocated by Stock and Yogo (2002) who suggest examining the improvement in model fit after including instruments (i.e., increase in F-value). As our monitoring variables are ordered and binary, the approach of Yogo and Stock is not possible and instead we relied on the approach suggested by Danaher et al. (2015) that advocates comparing models with and without instruments to assess if the Bayesian Information Criterion (BIC, which penalizes for model complexity) improves. For all three monitoring variables, we do indeed observe a decrease in BIC after including instruments. To verify the orthogonality assumption, we also re-estimated the final outcome model including the instruments, and found that none were significant ($p > .05$). In considering the validity of the instruments, it is important to note that abnormal returns result from investors' reactions to new information (i.e., the licensing agreement). The industry in which a licensee is located is known to investors.

Table 3
Correlation matrix and descriptive statistics.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Licensee market value	–														
2. Upstream process monitoring	0.037	–													
3. Downstream process monitoring	–0.093	0.631	–												
4. Downstream output monitoring	–0.055	0.075	0.073	–											
5. Licensing experience	–0.004	–0.066	–0.087	–0.058	–										
6. Partner-specific experience	–0.011	–0.059	0.013	–0.037	–0.021	–									
7. Investments	–0.051	0.260	0.360	0.142	0.054	0.010	–								
8. Product restrictions	–0.036	0.156	0.119	0.032	0.040	0.041	0.199	–							
9. Geographic restrictions	0.054	0.230	0.258	0.013	0.118	0.033	0.265	0.123	–						
10. Early-stage licensing	–0.109	0.349	0.294	0.046	0.134	0.018	0.293	0.114	0.140	–					
11. Foreign agreement	0.081	0.130	–0.015	–0.165	–0.084	–0.083	0.016	–0.007	0.012	–0.081	–				
12. Licensee R&D expenditures	–0.001	–0.079	–0.125	–0.032	–0.007	–0.037	–0.102	0.073	–0.105	–0.068	0.165	–			
13. Royalty-rate contract	0.159	0.164	0.211	0.061	0.051	–0.049	0.137	0.105	0.126	0.083	0.017	0.055	–		
14. Contract length	0.008	0.538	0.597	0.204	–0.059	–0.145	0.404	0.209	0.203	0.337	0.103	–0.115	0.303	–	
15. Licensee assets	–0.001	–0.076	–0.132	–0.033	–0.007	–0.038	–0.104	0.071	–0.107	–0.069	0.173	0.997	0.049	–0.126	–
Mean	0.003	0.596	0.949	0.404	8.912	0.110	0.574	0.728	0.338	0.301	0.426	8,037.807	0.750	9.131	67,850.570
Std. Dev.	0.091	0.855	0.991	0.493	88.611	0.433	0.685	0.447	0.475	0.461	0.496	51,323.890	0.435	0.861	430,465.300
Min	–0.300	0	0	0	0	0	0	0	0	0	0	0	0	5.852	0
Max	0.364	3	3	1	1033	4	2	1	1	1	1	543,937	1	10.906	4,546,723

N = 136, correlations over |0.169| are significant at $p < .05$, two-sided.

Third, when the three forms of monitoring are caused by common drivers, or when certain variables drive one form of monitoring as well as licensee market value, we might observe correlated errors between Equations 2–4. To address this issue, we use CMP's feature to specify correlated errors between the equations.

Finally, the fourth aspect is addressed by CMP's ability to simultaneously estimate systems of equations where dependent variables may be continuous, ordinal, or binary scaled as is required here. Given these considerations, we specify our model as follows:

$$\begin{aligned} \text{licensee market value}_i = & \beta_{10} + \beta_{11} \text{ upstream process monitoring}_i + \beta_{12} \text{ downstream process monitoring}_i \\ & + \beta_{13} \text{ downstream output monitoring}_i + \beta_{14} \text{ upstream process monitoring}_i \\ & * \text{licensing experience}_i + \beta_{15} \text{ downstream process monitoring}_i * \text{licensing experience}_i \\ & + \beta_{16} \text{ downstream output monitoring}_i * \text{licensing experience}_i \\ & + \beta_{17} \text{ upstream process monitoring}_i * \text{partner-specific experience}_i \\ & + \beta_{18} \text{ downstream process monitoring}_i * \text{partner-specific experience}_i \\ & + \beta_{19} \text{ downstream output monitoring}_i * \text{partner-specific experience}_i \\ & + \text{Control variables} + \omega^1_i \end{aligned} \quad (1)$$

$$\text{upstream process monitoring}_i = \beta_{20} + \text{Control variables}_i + \text{Instruments}_i + \omega^2_i \quad (2)$$

$$\text{downstream process monitoring}_i = \beta_{30} + \text{Control variables}_i + \text{Instruments}_i + \omega^3_i \quad (3)$$

$$\text{downstream output monitoring}_i = \beta_{40} + \text{Control variables}_i + \text{Instruments}_i + \omega^4_i \quad (4)$$

The vector *Control variables* includes licensing experience, partner-specific experience, investments, product restrictions, geographic restrictions, early-stage licensing, licensee R&D expenditures, a dummy for royalty-rate contract, the length of the contract, and licensee assets. The vector *Instruments* includes the aforementioned set of instruments. Equation 1 is specified with the dependent variable as a continuous variable, equations 2 and 3 as ordinal, and equation 4 is specified with a binary dependent variable.

4.5. Results and implications

Table 4 presents the results of our model estimation. Overall the results are supportive of our arguments and hypotheses. We start with the licensee market value equation (1). With regard to the main effects, while we do not find effects of upstream process monitoring ($\beta = -0.180$, $p > .05$) and downstream process monitoring on licensee market value ($\beta = 0.081$, $p > .05$), downstream outcome monitoring increases licensee market value ($\beta = 0.144$, $p < .01$).

We find strong support for all three interaction hypotheses with licensing experience (i.e. H1b, H2b, H3b). In line with H1b, upstream process monitoring increases licensee market value when the licensee has accumulated more licensing experience ($\beta = 0.019$, $p < .01$). In line with H2b, downstream process monitoring decreases licensee market value when the licensee has accumulated more licensing experience ($\beta = -0.005$, $p < .05$). Lastly, we find support for H3b, as the positive effect of downstream outcome monitoring on licensee market value is strengthened by licensing experience ($\beta = 0.007$, $p < .01$).

Interestingly, however, none of the interaction effects with partner-specific experience formulated in hypotheses H1a, H2a, and H3a are significant. In the discussion section, we will elaborate on the observed stark differences between licensing experience and partner-specific experience and draw conclusions for governance theory and managerial practice. Among the control variables, we only find that foreign agreements have a significant positive ($\beta = 0.089$, $p < .10$) impact on licensee market value.

We also gain some additional insight on what drives firms to include monitoring in licensing contracts. Specifically, when the licensee has accumulated more licensing experience, the licensing contract includes less process monitoring, both upstream ($\beta = -0.001$, $p < .01$) and downstream ($\beta = -0.001$, $p < .01$). In contrast, partner-specific experience (marginally) increases downstream process monitoring ($\beta = 0.202$, $p < .10$). Further, when licensing contracts are signed at early stages of technology development ($\beta = 0.417$, $p < .01$) and when licensees invest more in R&D ($\beta = 0.000$, $p < .01$), we observe more upstream process monitoring. Finally, more extensive licensing contracts, as proxied by the length of the contract, include more monitoring of all forms (upstream process monitoring: $\beta = 0.419$, $p < .01$; downstream process monitoring: $\beta = 0.576$, $p < .01$; downstream output monitoring: $\beta = 0.575$, $p < .01$).

4.6. Analyses of significant interactions

For a clearer interpretation of our hypotheses involving licensing experience, we estimated and plotted the simple slopes of upstream process monitoring, downstream process monitoring, and downstream output monitoring at no licensing experience and high licensing experience (1 SD above the mean), following Aiken and West (1991). Figs. 2–4 display the consequences of upstream process monitoring, downstream process monitoring, and downstream output monitoring, respectively, on licensee market value. For each type of monitoring, we observe a nonsignificant effect ($p > .05$) of monitoring

Table 4
Conditional Mixed Process Regression Estimates.

Variables	Licensee market value		Upstream process monitoring		Downstream process monitoring		Downstream output monitoring	
	β	S.E.	β	S.E.	β	S.E.	β	S.E.
Upstream process monitoring	–0.180	(0.164)						
Downstream process monitoring	0.081	(0.111)						
Downstream output monitoring	0.144	(0.032)						

Upstream process monitoring * licensing experience (H1b)	0.019	(0.005)						

Downstream process monitoring * licensing experience (H2b)	–0.005	(0.003)						
		**						
Downstream output monitoring * licensing experience (H3b)	0.007	(0.001)						

Upstream process monitoring * partner-specific experience (H1a)	0.027	(0.047)						
Downstream process monitoring * partner-specific experience (H2a)	–0.014	(0.033)						
Downstream output monitoring * partner-specific experience (H3a)	0.082	(0.058)						
<i>Control Variables</i>								
Licensing experience	0.000	(0.000)	–0.001	(0.000)	–0.001	(0.000)	0.000	(0.000)
				***		***		
Partner-specific experience	–0.021	(0.040)	–0.001	(0.133)	0.202	(0.103)*	–0.152	(0.338)
Investments	–0.036	(0.033)	–0.031	(0.117)	0.144	(0.114)	0.116	(0.144)
Product restrictions	0.034	(0.037)	0.064	(0.109)	–0.121	(0.146)	0.169	(0.279)
Geographic restrictions	0.030	(0.034)	0.153	(0.130)	0.181	(0.142)	0.076	(0.215)
Early stage licensing	0.026	(0.056)	0.417	(0.159)	0.205	(0.156)	–0.138	(0.216)

Foreign agreement	0.089	(0.049)*	0.200	(0.118)*	–0.066	(0.132)	–0.574	(0.195)

Licensee R&D expenditures	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)

Royalty-rate contract	0.032	(0.046)	0.052	(0.115)	0.096	(0.148)	0.298	(0.301)
Contract length	0.012	(0.045)	0.419	(0.095)	0.576	(0.100)	0.575	(0.179)
				***		***		***
Licensee assets	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)

SIC industry 38			–0.429	(0.158)	–0.510	(0.208)	0.441	(0.361)
				**		**		
SIC industry 73			–0.092	(0.143)	–0.421	(0.171)	0.568	(0.247)
						**		**
Constant	–0.224	(0.367)	–3.468	(0.810)	–4.314	(0.847)	–5.945	(1.629)
				***		***		***
Wald χ^2 (59) = 6667.53, Log pseudolikelihood = –200.424								

* $p < .10$, ** $p < .05$, *** $p < .01$, two-tailed. Note that the residuals of the upstream and downstream process monitoring equations are correlated ($t = 5.34$, $p < .01$) and the residuals of the upstream process monitoring and licensee market value equations are marginally correlated ($t = 1.82$, $p < .10$). All other residual correlations are insignificant.

when the licensee has no prior contracting experience (i.e., the left-hand side of Figs. 2–4). Yet, the market value effects of monitoring are all significant at high levels of licensing experience, demonstrating that licensing experience serves as a crucial boundary condition; all effects of monitoring on licensee market value are conditional on the licensee having accumulated a high level of licensing experience.

4.7. Additional tests

We undertook additional tests to assess the robustness of our results to assumptions regarding the estimator and interactive effects of monitoring and to examine the explanatory power of partner-specific experience.

Choice of estimator. Comparing the outcomes of standard OLS regression models (accounting for clustering) against CMP provides insight into the degree to which using CMP is critical to obtain the outcomes presented. The pattern of significance is very similar, which indicates that the substantive insights are not sensitive to the choice of estimator. We rely on CMP as this estimation procedure accounts for clustering, potential endogeneity, and correlated errors across the different equations.

Interactive effects between alternative forms of monitoring. To account for the possibility that the effects of monitoring are multiplicative, with alternative forms serving as complements or substitutes, we estimated new models including all possible two-way and three-way interaction effects. None of these additional interaction effects were significant ($p > .05$) and the reported results are robust.

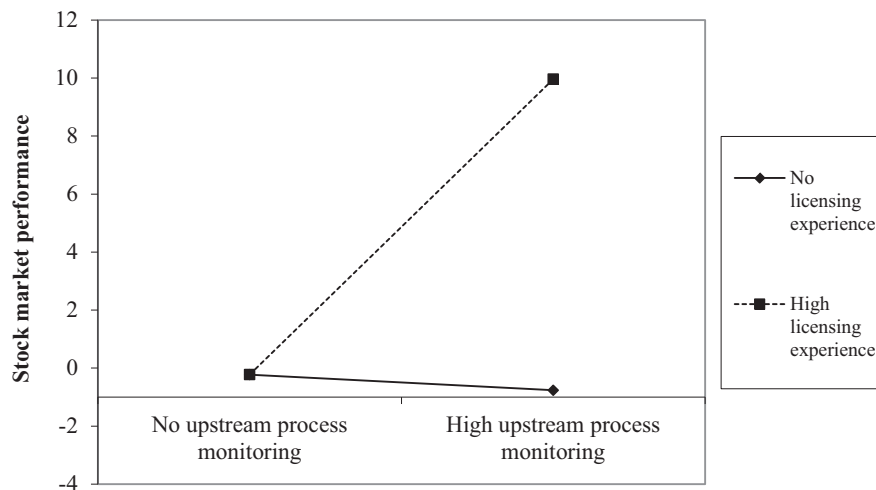


Fig. 2. Moderating effect of upstream process monitoring and licensing experience.

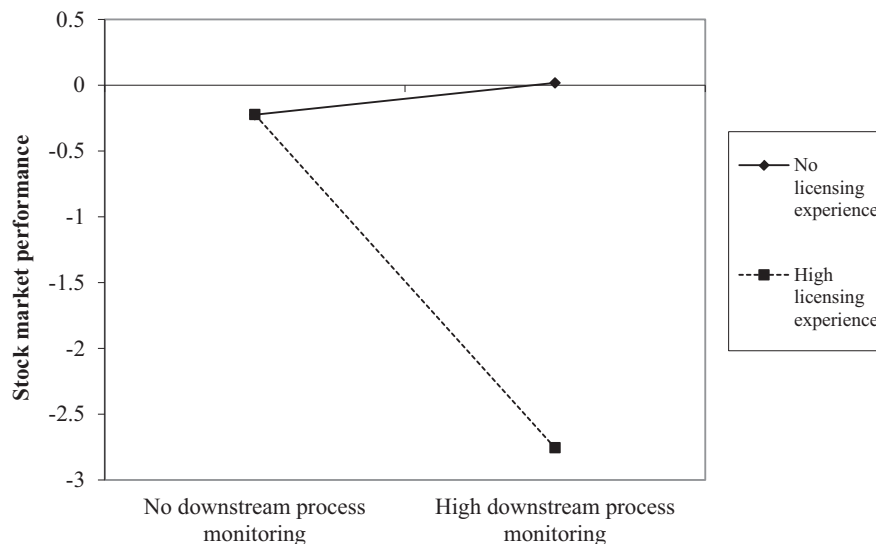


Fig. 3. Moderating effect of downstream process monitoring and licensing experience.

The value of adding interactions between partner-specific experience and monitoring. To consider the value of adding partner-specific experience interactions we considered a) the stability of our model without these interactions and b) incremental fit. Regarding stability, we estimated a model that did not include interactions between monitoring and partner-specific experience and found substantively similar results⁷. The effects of other variables, including interactions with licensing experience, were robust. Regarding incremental fit, we compared the full model (inclusive of interactions between monitoring and both licensing experience and partner-specific experience) with a model that excludes interactions with partner-specific experience. Adding interactions with partner-specific experience *deteriorates* model fit ($\Delta AIC = 4.3811$, $\Delta BIC = 16.0316$), indicating a general lack of explanatory power for partner-specific experience. We also compared the model that excludes only partner-specific experience with a model that excludes both partner-specific experience and licensing experience and found that adding interactions with licensing experience *improves* model fit ($\Delta AIC = -16.1962$, $\Delta BIC = -7.4582$). Since the results are robust across models, we reported the full model which allowed us to test all individual hypotheses and contrast licensing experience with (the more commonly studied) partner-specific experience.

⁷ Not tabulated here, but available from the authors.

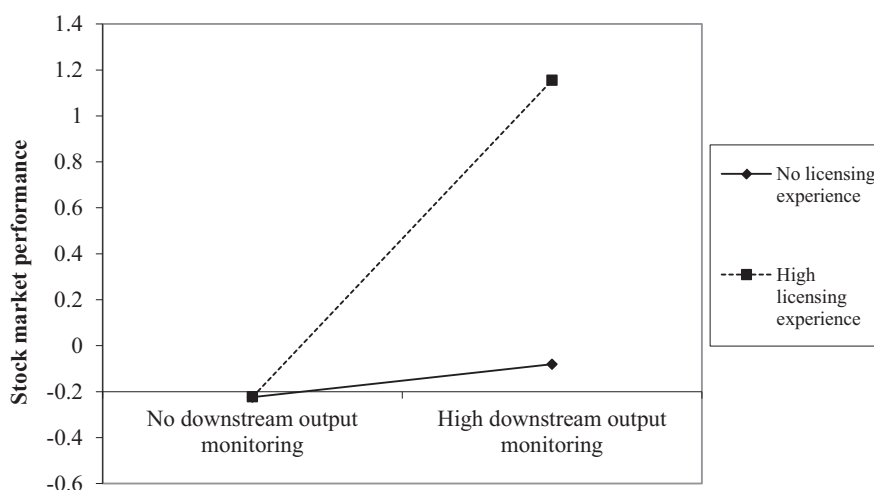


Fig. 4. Moderating effect of downstream output monitoring and licensing experience.

4.8. Theoretical implications

Licensing is a key form of exchange in technology markets but has received remarkably little attention in marketing. Via the development of novel theoretical arguments and the execution of an empirical test based on secondary data, we have taken several steps to address this void. Firstly, we extend the monitoring literature, premised on agency theory's two-sided moral hazard problem, by introducing the intuitive notion that a licensee can be the monitoring *and* the monitored party. Drawing on control and coordination arguments, we develop hypotheses linking upstream and downstream monitoring with licensee market value and we demonstrate that they exert starkly different effects. That the direction of monitoring is important, is underscored by the differential effects of downstream versus upstream process monitoring. Upstream process monitoring can create value for the licensee whereas downstream process monitoring reduces licensee firm value. Our findings also contribute to the broader inquiry in the governance literature into the differential impact of the various types of monitoring. Heide et al. (2007) found that monitoring can have different performance outcomes depending on the *type* of monitoring (output or process). Using a financial performance metric, to the best of our knowledge a first in the monitoring literature, and relying on first-hand information from contracts rather than survey-based reports, we replicate their finding and demonstrate that downstream output monitoring creates licensee firm value whereas downstream process monitoring negatively impacts licensee market value.

Secondly, we extend the literature on the effects of types of experience. Separating licensing experience (which is acquired through interactions with any partner) and partner-specific experience (which is acquired with the same exchange partner), we show that licensees benefit from upstream process monitoring and downstream output monitoring *conditional on* having accumulated licensing experience. Yet, licensing experience also makes licensees more vulnerable to downstream process monitoring: this form of monitoring hurts licensee market value, but only for experienced licensees. We do not find any such moderation effects with regard to partner-specific experience, further underscoring the need to distinguish between different sources of experience. Our overall finding that having prior experience is powerful when it is general (i.e., licensing experience) rather than when it is specific to the partner (partner-specific experience) contributes to the scant preexisting research on experience effects in the context of governance (e.g. Argyres & Mayer 2007; Vanneste & Puranam, 2010). The complexity and rapid technological change that characterizes technology licensing markets (John et al., 1999), makes technology licensing relatively discrete. The accumulation of greater licensing experience likely signals to investors that the licensee has developed higher-order licensing routines and capabilities, which can be subsequently used for monitoring purposes in the new licensing agreement. Experience with the same partner does not appear to have a similar signaling value. One explanation is that the pace of technological change makes partner-specific experiences less relevant, and partner-specific routines less portable. An alternative explanation is that a mismatch is created between the logic of appropriateness associated with repeated partnering and the logic of consequences associated with monitoring.

Third, the evidence we provide of governance's critical implications for licensee market value contributes to the governance literature at large. Unlike the vast majority of contracting research that focuses on intermediate constructs such as transaction costs or opportunism, our study is, to the best of our knowledge, the first to demonstrate that governance efforts at the transactional level explain firm financial performance. Our combined findings reflect that in their evaluation of the future revenue streams for licensees associated with licensing technologies, analysts and investors account for the contract including proper governance and the licensee having accumulated critical licensing experience in the past.

Fourth, the specification of monitoring equations provided additional theoretical insights. Licensing experience not only acts as a boundary condition of the effects of monitoring on licensee market value but also affects the ex-ante inclusion of monitoring in the licensing agreement. Since contractual specifications result from joint negotiations between licensee and licensor, licensing experience appears to impact both exchange parties. As licensees have greater licensing experience, process monitoring is reduced but not output monitoring. This suggests a shift in emphasis as licensing experience accumulates. The negative effect on process monitoring may result from the lesser need for monitoring as a coordination and communication tool when the licensee has greater experience. Licensing experience thus appears to tilt the governance balance towards contracts that deemphasize process control. From a theory point of view, the experience effects lend support to the (seldom tested) idea of “governance inseparability” (Argyres & Liebeskind, 1999) in that prior governance efforts impact the governance of a focal exchange. We demonstrate that there is such an effect of *having exchanged previously* with any partner on the governance of current exchanges, and their outcomes.

4.9. Managerial implications

We started our inquiry from the WIPO's (2016) explicit advice to include monitoring in licensing contracts to address uncertainty. Our research provides concrete guidance for licensees. Most importantly, we reject both the WIPO's and Anand and Khanna (2000) conclusion that licensing experience has no effect on the returns to licensing agreements. Even though we do not find *direct* effects of licensing experience, we find several significant *interaction* effects between licensing experience and monitoring. Investors appear to interpret experienced licensees' upstream process monitoring and downstream output monitoring in licensing contracts as signals of higher future revenue streams. The simple slope analyses clearly indicate that experience in licensing *greatly affects the returns of new licensing contracts*. The only dark side we uncover is that prior licensing experience also strengthens negative market value consequences associated with downstream process monitoring for licensees.

As pointed out earlier, we also observe that licensing contracts with more experienced licensees include significantly less process monitoring. While experienced licensees benefit from reduced downstream process monitoring, they *suffer* financially from reducing upstream process monitoring. The combination of a negative effect of licensing experience on upstream process monitoring and a positive interaction effect between licensing experience and upstream process monitoring on licensee market value appears to suggest that licensing experience can push licensees to draft imperfect contracts. Licensees can use the results presented here to broaden their thoughtworld on monitoring and to inform their decisions on the nature and direction of monitoring—accounting for their licensing experience accumulated in the past.

4.10. Limitations and issues for further research

Several of the findings raise new questions. First, our findings are suggestive of different contracting practices across countries. Foreign licensing agreements—with a non-US based party—include more upstream process monitoring but, somewhat surprisingly, less downstream output monitoring. Expanding the scope of inquiry to international licensing may help explain these empirical patterns. For example, the observed nuances may be caused by cultural, economic, regulatory, or other factors. Second, technology licensing success may be measured on other dimensions, such as speed to market which is important in sectors such as the pharmaceutical industry where a quick rollout of treatments of vaccines is crucial, or the licensee's success in terms of assimilating and productively utilizing (sometimes multiple) licensed technologies in innovative products. Our understanding of such market and product outcomes of technology licensing remains very limited, presenting interesting opportunities for future research. Third, future research may expand the scope of inquiry and focus on the performance of licensors, including their financial returns.

We are hopeful that our study will stimulate more research on licensing exchange, a unique form of exchange that requires theoretical extensions to existing governance theory. Given the growing economic importance of technology licensing, and its strategic importance for both licensors and licensees, the study of licensing exchange holds much promise for marketing academia and has great potential for novel practical insights.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Literature overview

	Upstream monitoring	Downstream monitoring	Output monitoring included	Process monitoring included
Dahlquist & Griffith, 2014	X	X	(X)*	(X)
Grewal et al., 2013		X	X	X
Wang, Li, Ross, & Craighead, 2013		X		X
Wang et al., 2013	X		(X)	(X)
Kashyap et al., 2012		X	X	X
Gilliland, Bello, & Gundlach, 2010	(X)	X		X
Heide et al., 2007	X		X	X
Murry & Heide, 1998		X		
Stump & Heide, 1996	X		X	
Agrawal & Lal, 1995		X	(X)	(X)
Noordewier, John, & Nevin, 1990	X		X	
This study	X	X	X	X

* (X) indicates that while the corresponding monitoring form is incorporated in the measures, the measurement approach does not allow to uniquely measure it (e.g. because of confoundment with another monitoring form).

We included empirical papers reporting on monitoring as a construct in an interfirm setting. We searched in the following journals from 1990–2018: *International Journal of Research in Marketing*, *Journal of Marketing*, *Journal of Marketing Research*, *Journal of the Academy of Marketing Science*, and *Marketing Science* (in alphabetical order).

Appendix B. Measurement items

Construct	Items/description/example (in Italics)	Theoretical and empirical source
Licensee market value	The difference between the expected and observed returns. Specifically, the cumulative average abnormal returns of the licensee on days –1 to +3.	• Raassens et al., 2012 • DataStream
Upstream process monitoring	3 items, coded each as 1 if present, 0 if absent. • Monitoring of the development process by the licensee towards the licensor. • Monitoring of the product and/or production process by the licensee towards the licensor. • Monitoring of the marketing, sales, advertising, and communication process by the licensee towards the licensor. Contract between Activision (publisher) and licensor Sony (SCEA): <i>“SCEA will inform Publisher as to the status of the Product Proposal and Review Process for such Licensed Product.”</i>	• Ryall and Sampson (2006) • CORI
Downstream process monitoring	3 items, coded each as 1 if present, 0 if absent. • Monitoring of the development process by the licensor towards the licensee. • Monitoring of the product and/or production process by the licensor towards the licensee. • Monitoring of the marketing, sales, advertising, and communication process by the licensor towards the licensee. Contract between K-V Pharmaceutical Company and licensor Femme-Pharma: <i>“Licensee shall provide reasonable advance notice to licensor of</i>	• Ryall and Sampson (2006) • CORI

(continued on next page)

Measurement items (continued)

Construct	Items/description/example (in Italics)	Theoretical and empirical source
	<i>all communications provided by licensee's sales representatives for the positioning, promotion or selling messages for the Products for licensors review and approval".</i>	
Downstream output monitoring	1 item, coded as 1 if the right to monitor financial results by the licensor is present, 0 else.	• Ryall and Sampson (2006) • CORI
Investments	3 items, coded each as 1 if present, 0 if absent. Does the license require investments by the licensee in terms of: • R&D investment • Manufacturing investment • Marketing investment Contract between licensee Rycor Technology and licensor Autoimmune: "licensee shall within forty-five (45) days after each calendar quarter furnish to licensor a written quarterly report showing: (i) the gross sales of the Licensed Products ... the calculation of Net Sales from such gross sales; (ii) the royalty due thereon; (iii) withholding taxes, if any, required by law to be deducted in respect of such royalties; and (iv) the exchange rates ... If no royalty is due for any royalty period hereunder, licensee shall so report. Licensee shall keep complete and accurate records in sufficient detail to properly reflect all gross sales and Net Sales."	• Somaya et al. (2010) • CORI
Product restrictions	1 item, coded as 1 if license contains product use restrictions, 0 else.	• Somaya et al. (2010) • CORI
Geographic restrictions	1 item coded as 1 if the license contains geographical restrictions, 0 else.	• Somaya et al. (2010) • CORI
Early-stage licensing	1 item coded as 1 if the licensee is dependent on the licensor for further-development of the licensed technology after contract signing, 0 else.	• Based on Chen and Bharadwaj (2009). • CORI
Foreign agreement	1 item coded as 1 if the exchange is with a non-US based party, 0 else.	• CORI
Licensing experience	The number of prior licensee agreements of the licensee (with any partner). Log transformed	• SDC Platinum
Partner-specific experience	The number of prior licensee agreements signed by the licensee (with the same partner).	• SDC Platinum
Licensee R&D expenditures	The percentage of turnover spent on R&D.	• DataStream
Royalty-rate contract	1 item coded as 1 the contract is royalty-based, 0 else.	• CORI
Contract length	Contract length is the total word count of the contract.	• CORI
Licensee assets	Measured as the total \$ financial assets of the licensee.	• DataStream
Industry dummies	1 if the licensee is assigned to SIC code 38 (0 else) 1 if the licensee is assigned to SIC code 73 (0 else)	• CORI

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